INVENTORY MANAGEMENT SYSTEM

PROJECT NO. 19 • PYTHON 3 • TEST CASES & EXPECTED OUTPUT

Purpose: Verify Add, Update, Delete, Search/View, stock value, low-stock report, and input validation.

TEST CASE 1 — NORMAL INPUT (ADD ITEM)

INPUT / ACTION	EXPECTED OUTPUT
Menu option: 1 (Add) Name: Pen Quantity: 50 Price: 10	Item added. ID: I101 | Pen | Qty: 50 | Price: 10

Note: First item ID must be I101.

TEST CASE 2 — BOUNDARY / MINIMUM INPUT

INPUT / ACTION	EXPECTED OUTPUT
Menu option: 1 (Add) Name: Eraser Quantity: 0 Price: 5	Item added. ID: I102 | Eraser | Qty: 0 | Price: 5 Low stock alert: Eraser (0)

Note: Quantity 0 is valid; low stock means quantity below 5.

TEST CASE 3 — INVALID INPUT

INPUT / ACTION	EXPECTED OUTPUT
(a) Add: Pencil, Quantity: -5, Price: 4 (b) Option 3 (Update), ID: I999	(a) Invalid quantity. Quantity must be a non-negative whole number. Item not added. (b) Item ID I999 not found.

Note: Invalid item should not be added; inventory remains unchanged.

TEST CASE 4 — DIFFERENT VALID SCENARIO

INPUT / ACTION	EXPECTED OUTPUT
Inventory before test: I101 Pen (Qty 50 × Price 10) I102 Notebook (Qty 3 × Price 40) (a) Option 5: Total stock value (b) Option 6: Low stock report (c) Option 4: Delete I101	(a) Pen: 500 | Notebook: 120 Total stock value: 620 (b) Low stock (below 5): I102 | Notebook | Qty: 3 (c) Item 'Pen' deleted. Items left: 1

Note: For this case I102 must be Notebook (3 × 40). If Test Case 2 added Eraser, update it before this scenario.

TEST EXECUTION CHECKLIST

#	Check	Pass / Fail
1	Add Pen; verify ID I101 and details	_____
2	Add Eraser with quantity 0; verify low-stock alert	_____

3	Reject negative quantity and missing ID I999	_____
4	Verify total 620, low-stock report, and delete result	_____

Project: Inventory Management System **Language:** Python 3